MATHEMATICS

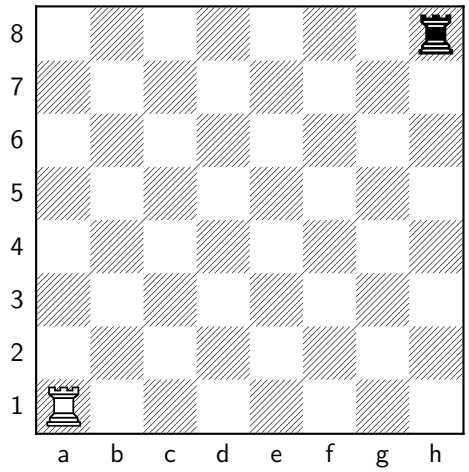
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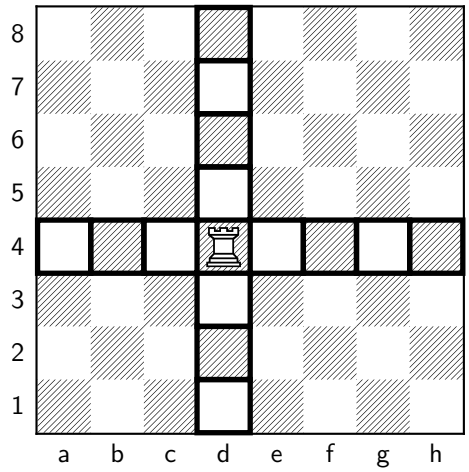
Spring 2025 Math Problem of the Week: Problem #6

Due noon Monday, February 24
Open to all UNA undergraduates

Directions: Deliver your written solution to me (Dr. Bryant) in CPTMTH 328 by noon Monday, February 24 or, if you cannot find me, you may leave it with Ms. Buckley in CPTMTH 377. All work must be your own. You are required to justify your response fully. If you utilize a well-known theorem, state the theorem's full name and proposition, showing that the conditions of the theorem are met. Good luck!

Problem #6: Consider the following game: Two rooks of opposite colors are placed on squares a1 and h8 of a chessboard as pictured below on the left. Beginning with white, each rook will take turns making random, equally likely, legal chess moves (any distance horizontally or vertically as pictured on the right). No rook may remain stationary on its turn. If a rook can land on the same square as its counterpart, it must do so on that turn, capturing the opponent and winning the game.





What is the expected number of turns until the game ends? What is the probability that the white rook is the winner?